



6572 - Constraining the Birth Conditions of Super Star Clusters in Henize 2-10

Cycle: 3, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	Henize 2-10	MIRI Medium Resolution Spectroscopy	(2) HENIZE2-10
	51	Henize 2-10	MIRI Medium Resolution Spectroscopy	(2) HENIZE2-10
	2	Background	MIRI Medium Resolution Spectroscopy	(3) BACKGROUND
	52	Background	MIRI Medium Resolution Spectroscopy	(3) BACKGROUND

ABSTRACT

Henize2-10 is a nearby compact dwarf galaxy with extreme star formation and numerous super star clusters. Super Star Clusters (SSCs) are thought to be present day analogs of globular clusters. The SSCs are massive compact clusters of stars, but the physical conditions of a star-forming region that promote such clusters are fundamentally not understood.

We have gathered continuum data from 5--350GHz to characterize embedded natal SSCs in He2-10 and their ultra dense (UD) HII regions. Critically missing, however, are high resolution nebular diagnostics of the physical and geometric properties of these natal environments. We propose VLA observations of 16 Hydrogen radio recombination lines at 9 and 22GHz to determine the electron density, electron temperature, and size of these UD HII regions that host natal SSCs.

JWST Proposal 6572 (Created: Wednesday, May 21, 2025, 3:00:08PM Eastern Standard Time) - Overview

We propose joint JWST proposals to measure the line emission ratios $[\text{NeIII}]/[\text{NeII}]$, $[\text{ArIII}]/[\text{ArII}]$, and $[\text{SIV}]/[\text{SIII}]$, which we will be able to model the ionization parameter and the radiation hardness. The ionization parameter is the ratio of ionizing photon density to electron density and is directly related to the ratio of radiation pressure to gas pressure. With these, we can probe internal morphology of the embedded HII regions and provide insight into the SSC assembly process.

This is a joint NRAO JWST request for proposal VLA/24B-281.

OBSERVING DESCRIPTION

We propose to observe the dwarf galaxy Henize 2-10 (Hen 2-10) with MIRI MRS with all three gratings to get continuous spectral coverage from 5 to 28 microns. The spatial resolution at the distance of Hen 2-10 ($D = 9$ Mpc) varies from 16 to 49 pc, which is sufficient to resolve the environment surrounding individual natal super star clusters, even at the longest wavelengths. With a mosaic of two pointings, we are able to cover the entire star forming region of Hen 2-10. Since the galaxy fills the entire mosaic, we require an off-galaxy background frame. Exposure times were calculated using a template spectrum from Spitzer observations of 30 Doradus scaled to the distance of Hen 2-10, resulting in a total exposure time of 2.64 hours and a total charged time of 5.62 hours. This is a joint with VLA/24B-281, with the NRAO being the primary observatory.

Proposal 6572 - Targets - Constraining the Birth Conditions of Super Star Clusters in Henize 2-10

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(2)	HENIZE2-10	RA: 08 36 15.1688 (129.0632033d) Dec: -26 24 34.10 (-26.40947d) Equinox: J2000	Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Blue compact dwarf galaxies, Starburst galaxies] Extended=YES				
Fixed Targets	(3)	BACKGROUND	RA: 08 36 15.0000 (129.0625000d) Dec: -26 23 40.00 (-26.39444d) Equinox: J2000		
	Comments: Category=Calibration Description=[Telescope/sky background]				

Proposal 6572 - Observation 1 - Constraining the Birth Conditions of Super Star Clusters in Henize 2-10

Observation	Proposal 6572, Observation 1: Henize 2-10												Wed May 21 20:00:08 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[Background (Obs 2)]												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(2)	HENIZE2-10	RA: 08 36 15.1688 (129.0632033d) Dec: -26 24 34.10 (-26.40947d) Equinox: J2000				Epoch of Position: 2015.5						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=Galaxy Description=[Blue compact dwarf galaxies, Starburst galaxies] Extended=YES												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID				
	1	2 HENIZE2-10	F1000W	FAST	4	1	1	11.1	144503.13				
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		YES			FULL			Allow Auto Reorder				
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order		
	1	2	10.0		10.0		0.0		0.0		DEFAULT		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
	2	2-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F770W	SLOWR1	6	1	1	Dither 1	4	4	573.358	
	1	LONG(C)	MRSLONG		FASTR1	125	1	1	Dither 1	4	4	1387.52	144503.9
	1	LONG(C)	MRSSHORT		FASTR1	125	1	1	Dither 1	4	4	1387.52	144503.3
	2		IMAGER	F770W	SLOWR1	6	1	1	Dither 1	4	4	573.358	
	2	MEDIUM(B)	MRSLONG		FASTR1	140	1	1	Dither 1	4	4	1554.022	144503.8
	2	MEDIUM(B)	MRSSHORT		FASTR1	140	1	1	Dither 1	4	4	1554.022	144503.2
	3		IMAGER	F770W	SLOWR1	6	1	1	Dither 1	4	4	573.358	
	3	SHORT(A)	MRSLONG		FASTR1	77	1	1	Dither 1	4	4	854.712	144503.7
	3	SHORT(A)	MRSSHORT		FASTR1	77	1	1	Dither 1	4	4	854.712	144503.4

Proposal 6572 - Observation 1 - Constraining the Birth Conditions of Super Star Clusters in Henize 2-10

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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Proposal 6572 - Observation 51 - Constraining the Birth Conditions of Super Star Clusters in Henize 2-10

Observation	Proposal 6572, Observation 51: Henize 2-10												Wed May 21 20:00:08 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[Background (Obs 52)]													
Diagnostics	(Visit 51:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(2)	HENIZE2-10	RA: 08 36 15.1688 (129.0632033d)				Epoch of Position: 2015.5							
			Dec: -26 24 34.10 (-26.40947d)											
			Equinox: J2000											
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Blue compact dwarf galaxies, Starburst galaxies] Extended=YES												
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
	F1000W	All MRS				YES				FULL		Allow Auto Reorder		
Mosaic	Rows	Columns	Row Overlap %			Column Overlap %			Row shift (deg)		Column shift (deg)		Tile Order	
	1	2	10.0			10.0			0.0		0.0		DEFAULT	
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
	2	2-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	SLOWR1	6	1	1	Dither 1	4	4	573.358		
	1	LONG(C)	MRSLONG		FASTR1	125	1	1	Dither 1	4	4	1387.52	144503.9	
	1	LONG(C)	MRSSHORT		FASTR1	125	1	1	Dither 1	4	4	1387.52	144503.3	
	2		IMAGER	F770W	SLOWR1	6	1	1	Dither 1	4	4	573.358		
	2	MEDIUM(B)	MRSLONG		FASTR1	140	1	1	Dither 1	4	4	1554.022	144503.8	
	2	MEDIUM(B)	MRSSHORT		FASTR1	140	1	1	Dither 1	4	4	1554.022	144503.2	
	3		IMAGER	F770W	SLOWR1	6	1	1	Dither 1	4	4	573.358		
	3	SHORT(A)	MRSLONG		FASTR1	77	1	1	Dither 1	4	4	854.712	144503.7	
	3	SHORT(A)	MRSSHORT		FASTR1	77	1	1	Dither 1	4	4	854.712	144503.4	

Proposal 6572 - Observation 51 - Constraining the Birth Conditions of Super Star Clusters in Henize 2-10

Special Requirements	Sequence Observations 51, 52, Non-interruptible
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Proposal 6572 - Observation 2 - Constraining the Birth Conditions of Super Star Clusters in Henize 2-10

Observation	Proposal 6572, Observation 2: Background												Wed May 21 20:00:08 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [Henize 2-10 (Obs 1)]													
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(3)	BACKGROUND	RA: 08 36 15.0000 (129.0625000d) Dec: -26 23 40.00 (-26.39444d) Equinox: J2000											
	Comments: Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID					
	1	2 HENIZE2-10	F1000W	FAST	4	1	1	11.1	144503.13					
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
	2	2-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	SLOWR1	6	1	1	Dither 2	2	2	286.679		
	1	LONG(C)	MRSLONG		FASTR1	125	1	1	Dither 2	2	2	693.76	144503.9	
	1	LONG(C)	MRSSHORT		FASTR1	125	1	1	Dither 2	2	2	693.76	144503.3	
	2		IMAGER	F770W	SLOWR1	6	1	1	Dither 2	2	2	286.679		
	2	MEDIUM(B)	MRSLONG		FASTR1	140	1	1	Dither 2	2	2	777.011	144503.8	
	2	MEDIUM(B)	MRSSHORT		FASTR1	140	1	1	Dither 2	2	2	777.011	144503.2	
	3		IMAGER	F770W	SLOWR1	6	1	1	Dither 2	2	2	286.679		
	3	SHORT(A)	MRSLONG		FASTR1	77	1	1	Dither 2	2	2	427.356	144503.7	
	3	SHORT(A)	MRSSHORT		FASTR1	77	1	1	Dither 2	2	2	427.356	144503.4	

Proposal 6572 - Observation 2 - Constraining the Birth Conditions of Super Star Clusters in Henize 2-10

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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Proposal 6572 - Observation 52 - Constraining the Birth Conditions of Super Star Clusters in Henize 2-10

Observation	Proposal 6572, Observation 52: Background												Wed May 21 20:00:08 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [Henize 2-10 (Obs 51)]												
Diagnostics	(Visit 52:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(3)	BACKGROUND	RA: 08 36 15.0000 (129.0625000d) Dec: -26 23 40.00 (-26.39444d) Equinox: J2000										
	Comments: Category=Calibration Description=[Telescope/sky background]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F1000W	All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
	2	2-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F770W	SLOWR1	6	1	1	Dither 2	2	2	286.679	
	1	LONG(C)	MRSLONG		FASTR1	125	1	1	Dither 2	2	2	693.76	144503.9
	1	LONG(C)	MRSSHORT		FASTR1	125	1	1	Dither 2	2	2	693.76	144503.3
	2		IMAGER	F770W	SLOWR1	6	1	1	Dither 2	2	2	286.679	
	2	MEDIUM(B)	MRSLONG		FASTR1	140	1	1	Dither 2	2	2	777.011	144503.8
	2	MEDIUM(B)	MRSSHORT		FASTR1	140	1	1	Dither 2	2	2	777.011	144503.2
	3		IMAGER	F770W	SLOWR1	6	1	1	Dither 2	2	2	286.679	
	3	SHORT(A)	MRSLONG		FASTR1	77	1	1	Dither 2	2	2	427.356	144503.7
	3	SHORT(A)	MRSSHORT		FASTR1	77	1	1	Dither 2	2	2	427.356	144503.4

Proposal 6572 - Observation 52 - Constraining the Birth Conditions of Super Star Clusters in Henize 2-10

Special Requirements	Sequence Observations 51, 52, Non-interruptible
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